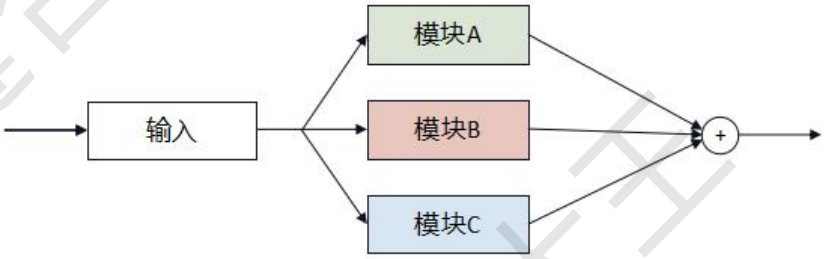
多尺度特征提取，是我们论文中非常常见的一个东西，我们进行特征提取时，通常会提到这一点。

之前分享的那个 xx-former 并行例子，SG-Former，A+B+C，并行缝合了 Vit 、Swin 、PVT 中提取的注意力机制，当作 token mixer，结合了三者不同的提取特征的能力，其实也是进行多尺度混合特征提取。

(ICCV2023) SG-Former: Self-guided Transformer with Evolving Token Reallocation



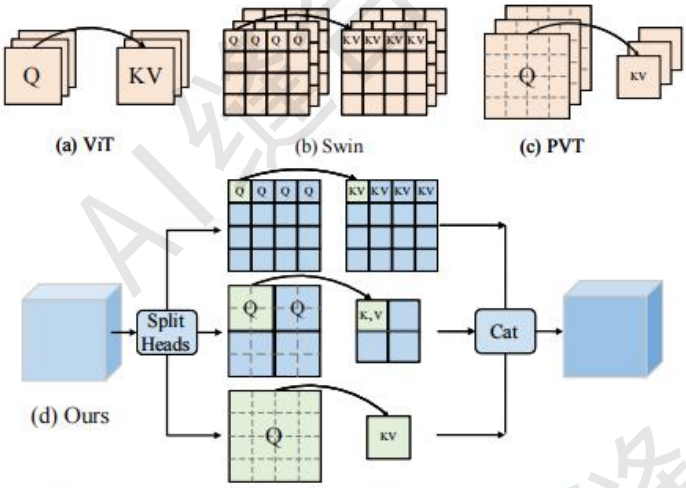


Figure 5. Comparison of Local Global information and hybrid-scale. (A) Self-attention in ViT is single scale and has huge computation cost, thereby applicable to only small feature maps. (B) Self-attention in Swin Transformer is single scale and is limited within small windows, which reduces computation cost and preserves fine-grained information; however, the attention is local. (C) Self-attention in PVT is single scale. It reduces tokens alongside computation cost, and preserves global receptive field; however, it captures coarse information only. (D) Our Self-attention is hybrid-scale and performs attention from local to global areas. It reduces computation cost, preserves global receptive field, and captures fine-grained hybrid-scale information.

（Arxiv 2023） MMViT: Multiscale Multiview Vision Transformers

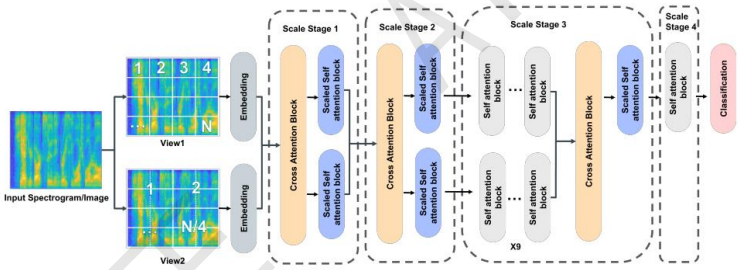


Figure 1: *An overview of the 16-block MMViT model. The input spectrogram is first patchified into two views with different resolutions. Both views are then encoded with a learnable spatio-temporal positional encoding layer. The embedded features pass through four scale stages, each with multiple transformer blocks. The second-to-last block of each stage except the last stage is used to fuses the information from both views using cross-attention, while the last block uses scaled self-attention to reduce spatio-temporal resolution and increase feature channels. The other blocks use regular self-attention.*

MMViT 这篇文章，分为俩个分支，我们设计模型的时候，可以借鉴这种方式 ，就比如使用窗口注意力机制，分为两个分支，每个分支窗口大小设置不一样。这样多尺度不就来了，不就可以水了。

双分支后面需要融合模块，我们使用 Cross Attention，也可以使用其他的即插即用的特征融合模块。

（TMM2023) DilateFormer: Multi-Scale Dilated Transformerfor Visual Recognition

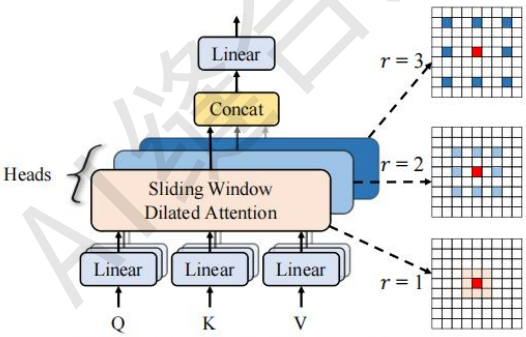


Fig. 4: **Illustration of Multi-Scale Dilated Attention (MSDA).** First, the channels of the feature map are split into different heads. Then, the self-attention operation is performed among the colored patches in the window surrounding the red query patch, using different dilation rates in different heads. Besides, features in different heads are concatenated together and then fed into a linear layer. By default, we use a 3×3 kernel size with dilation rates $r = 1$, 2 and 3, and the sizes of attended receptive fields in different heads are 3×3, 5×5 and 7×7.

不同于 MMVIT 对分支窗口大小设置不同，DilateFormer 对不同的注意头采用不同大小的膨胀卷积，这也是一种多尺度的方式。

除此之外，我们使用不同大小的卷积，或者不同的模块、注意力机制、fft 等，我们写论文的时候，都可以写，我们的模块结合了 xxx 不同的提取特征的能力，对图像进行多尺度特征提取。